

MAD & PWA Lab

Journal

Experiment No.	11
Experiment Title.	To use google Lighthouse PWA Analysis Tool to test the PWA functioning.
Roll No.	58
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Class	D15A
Subject	MAD & PWA Lab
Lab Outcome	LO6: Develop and Analyze PWA Features and deploy it over app hosting solution
Grade:	

Experiment . 11

Aim:

To analyze the functionality and performance of a Progressive Web App (PWA) using the Google Lighthouse PWA Analysis Tool.

Theory:

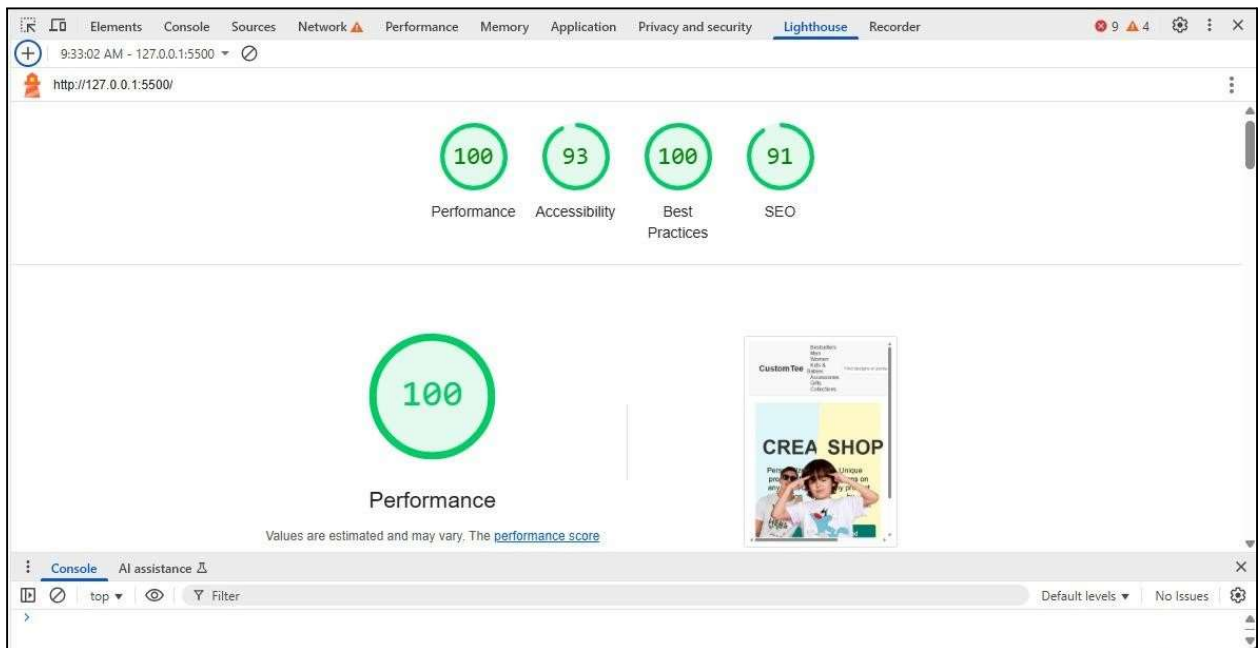
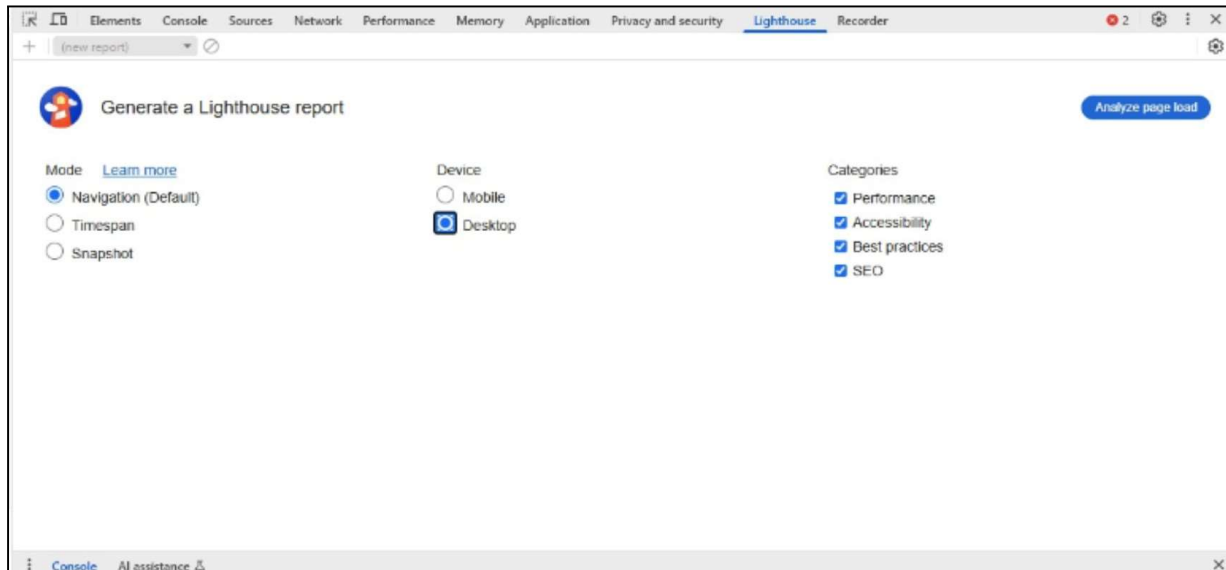
Google Lighthouse is an open-source, automated auditing tool developed by Google that assesses the quality of web pages. It performs audits across several categories including Performance, Accessibility, Best Practices, SEO, and Progressive Web App (PWA) compliance.

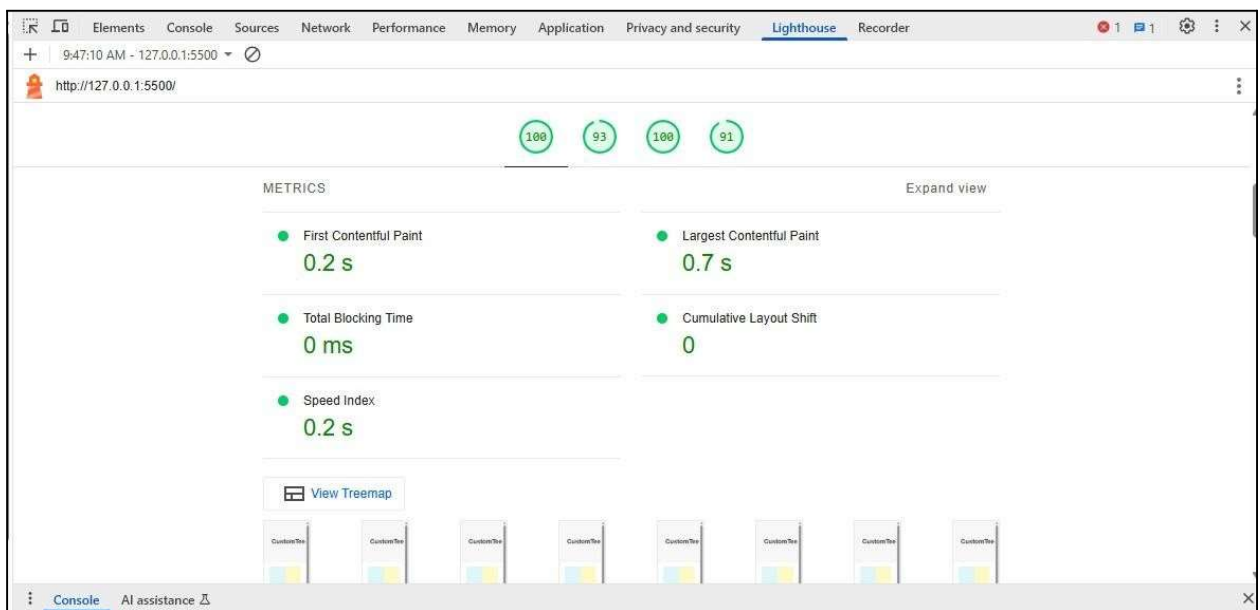
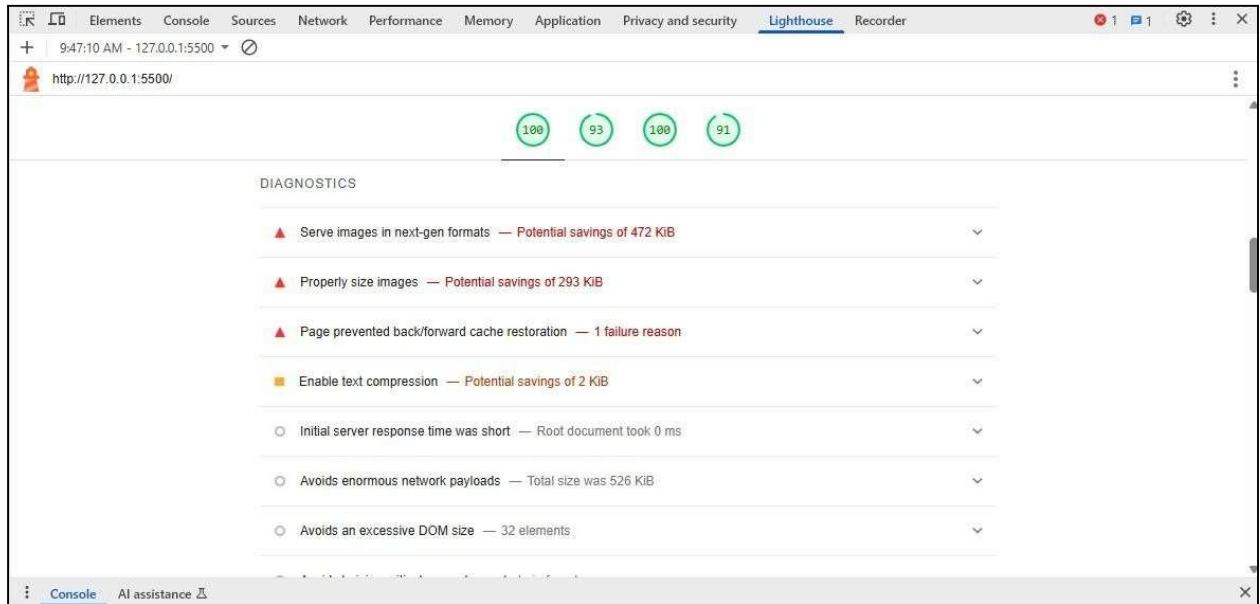
For PWAs, Lighthouse ensures that the application follows modern web standards such as:

- Presence and configuration of a **Service Worker** for offline capabilities
- A well-defined **Web App Manifest** with appropriate metadata (icons, theme color, display mode)
- **Responsive design** and proper **viewport configuration** for mobile compatibility
- **Installability** on user devices
- Offline functionality and adherence to **Google's baseline PWA checklist**

Additionally, Lighthouse scores the site on performance (loading speed and responsiveness), accessibility (support for screen readers and assistive tools), and web development best practices (secure connections, modern syntax, deprecated feature avoidance, etc.).

Output:





Conclusion:

Thus, we successfully used the Google Lighthouse PWA Analysis Tool to audit and verify the Progressive Web App functionality of our project. The report provided insights into areas of improvement, ensuring that our PWA meets modern standards for performance, accessibility, and installability across devices.

